

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

## Chapter 6: Triangles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

### 1 Basic Proportionality Theorem

1. In  $\triangle ABC$ ,  $D$  lies on  $AB$  and  $E$  lies on  $AC$  with  $DE \parallel BC$ . If  $AD = x$ ,  $DB = x - 2$ ,  $AE = x + 2$  and  $EC = x - 1$ , find  $x$ . **CBSE 2026, 2 marks**

SOLUTION

By the Basic Proportionality Theorem,  $\frac{AD}{DB} = \frac{AE}{EC}$ .

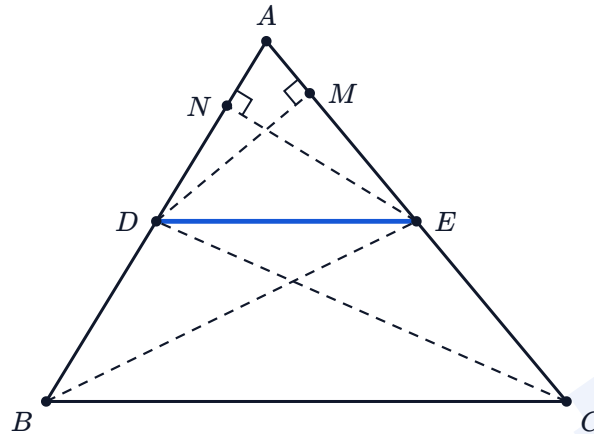
$$\frac{x}{x-2} = \frac{x+2}{x-1}, \text{ so } x(x-1) = (x+2)(x-2)$$
$$x^2 - x = x^2 - 4$$

**Answer:**  $x = 4$

2. State the Basic Proportionality Theorem and prove it. **CBSE 2026, 5 marks**

**SOLUTION**

**Statement:** If a line is drawn parallel to one side of a triangle and cuts the other two sides at distinct points, it divides those two sides in the same ratio.



**Given:**  $\triangle ABC$  with  $DE \parallel BC$ , where  $D$  is on  $AB$  and  $E$  is on  $AC$ . **To prove:**  $\frac{AD}{DB} = \frac{AE}{EC}$ .

**Construction:** Join  $BE$  and  $CD$ . Draw  $EN \perp AB$  and  $DM \perp AC$ .

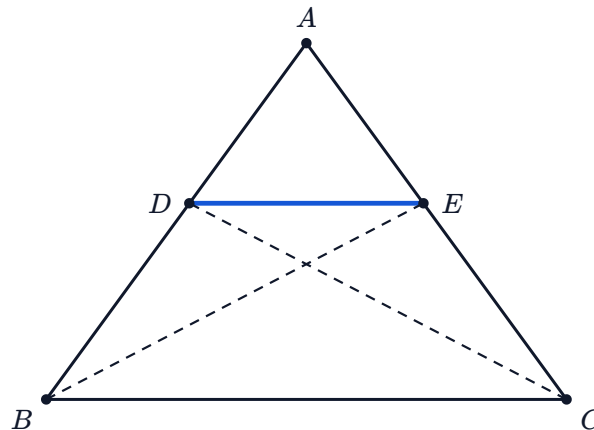
$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB} \dots (1)$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CED)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC} \dots (2)$$

$\triangle BDE$  and  $\triangle CED$  stand on the same base  $DE$  and lie between the same parallels  $DE$  and  $BC$ , so  $\text{ar}(\triangle BDE) = \text{ar}(\triangle CED) \dots (3)$

From (1), (2) and (3):  $\frac{AD}{DB} = \frac{AE}{EC}$ . **Proved.**

3. Prove that a line drawn parallel to one side of a triangle, cutting the other two sides at distinct points, divides those two sides in the same ratio. **CBSE 2024, 5 marks**



**SOLUTION**

Given: in  $\triangle ABC$ ,  $DE \parallel BC$ , with  $D$  on  $AB$  and  $E$  on  $AC$ . To prove:  $\frac{AD}{DB} = \frac{AE}{EC}$ .

Construction: join  $BE$  and  $CD$ . Draw  $EN \perp AB$  and  $DM \perp AC$ .

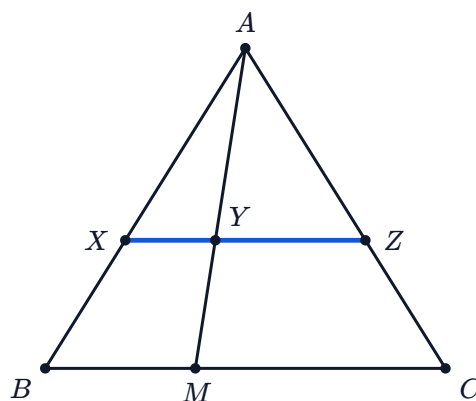
$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB}$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CED)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}$$

$\triangle BDE$  and  $\triangle CED$  are on the same base  $DE$  and between the same parallels  $DE$  and  $BC$ , so  $\text{ar}(\triangle BDE) = \text{ar}(\triangle CED)$ .

Hence  $\frac{AD}{DB} = \frac{AE}{EC}$ .

4. In  $\triangle ABC$ ,  $X$  lies on  $AB$  and  $Z$  on  $AC$  with  $XZ \parallel BC$ .  $M$  is a point on  $BC$ , and  $AM$  meets  $XZ$  at  $Y$ . If  $AZ = 3$  cm,  $ZC = 2$  cm,  $BM = 3$  cm and  $MC = 5$  cm, find  $XY$ . **CBSE 2023, 2 marks**



**SOLUTION**

$$XZ \parallel BC, \text{ so } \frac{AX}{AB} = \frac{AZ}{AC} = \frac{3}{3+2} = \frac{3}{5}.$$

In  $\triangle ABM$ ,  $XY \parallel BM$ , so  $\triangle AXY \sim \triangle ABM$  (AA).

$$\frac{XY}{BM} = \frac{AX}{AB} = \frac{3}{5}, \text{ so } XY = \frac{3}{5} \times 3$$

**Answer:**  $XY = 1.8$  cm

5. A line parallel to one side of a triangle meets the other two sides at distinct points. Prove that it divides those two sides in the same ratio. **CBSE 2023, 5 marks**

**SOLUTION**

This is the Basic Proportionality Theorem. In  $\triangle ABC$ , let  $DE \parallel BC$  with  $D$  on  $AB$  and  $E$  on  $AC$ . Join  $BE$  and  $CD$ , and draw  $EN \perp AB$  and  $DM \perp AC$ .

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB} \text{ and } \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CED)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}$$

$\triangle BDE$  and  $\triangle CED$  are on the same base  $DE$  and between the same parallels, so their areas are equal.

**Hence**  $\frac{AD}{DB} = \frac{AE}{EC}$ . **Proved.** (The figure is with the 2026 question above.)

## 2 Criteria for similarity of triangles

6.  $\triangle ABC \sim \triangle DEF$  with  $2AB = DE$  and  $BC = 8$  cm. Find  $EF$ . (a) 4 cm (b) 8 cm (c) 12 cm (d) 16 cm **CBSE 2026, 1 mark**

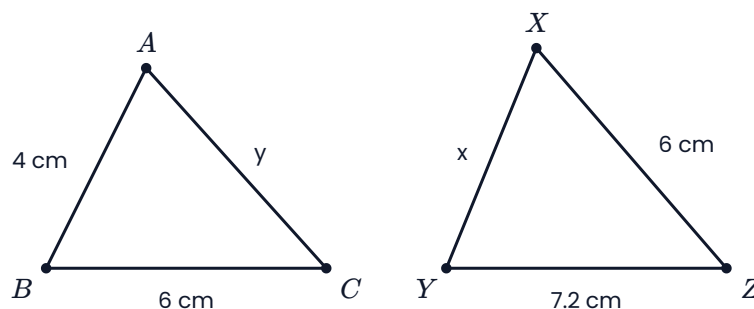
**SOLUTION**

Corresponding sides are proportional:  $\frac{AB}{DE} = \frac{BC}{EF}$ .

$$\frac{AB}{DE} = \frac{1}{2}, \text{ so } \frac{8}{EF} = \frac{1}{2} \text{ and } EF = 16 \text{ cm.}$$

**Answer: (d)**

7.  $\triangle ABC \sim \triangle XYZ$  with  $AB = 4$  cm,  $BC = 6$  cm,  $AC = y$  cm,  $XY = x$  cm,  $YZ = 7.2$  cm and  $XZ = 6$  cm. Find  $x$  and  $y$ . **CBSE 2026, 2 marks**



**SOLUTION**

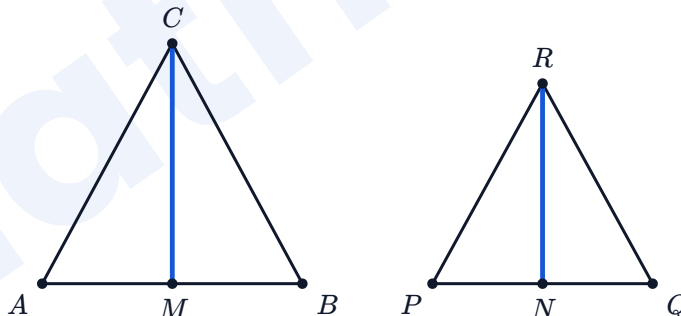
$$\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ}, \text{ and } \frac{BC}{YZ} = \frac{6}{7.2} = \frac{5}{6}$$

$$\frac{4}{x} = \frac{5}{6}, \text{ so } x = \frac{24}{5} = 4.8$$

$$\frac{y}{6} = \frac{5}{6}, \text{ so } y = 5$$

**Answer:**  $x = 4.8, y = 5$

8.  $CM$  and  $RN$  are medians of  $\triangle ABC$  and  $\triangle PQR$ , and  $\triangle ABC \sim \triangle PQR$ . Prove that (i)  $\triangle AMC \sim \triangle PNR$  (ii)  $\triangle CMB \sim \triangle RNQ$ . **CBSE 2026, 5 marks**



**SOLUTION**

$$\triangle ABC \sim \triangle PQR, \text{ so } \angle A = \angle P, \angle B = \angle Q \text{ and } \frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP}.$$

$M$  and  $N$  are mid-points, so  $AB = 2AM = 2MB$  and  $PQ = 2PN = 2NQ$ . Hence  $\frac{AM}{PN} = \frac{MB}{NQ} = \frac{AB}{PQ}$ .

(i) In  $\triangle AMC$  and  $\triangle PNR$ :  $\frac{AM}{PN} = \frac{AC}{PR}$  and the included angles  $\angle A = \angle P$ . **So**  $\triangle AMC \sim \triangle PNR$  (**SAS**).

(ii) In  $\triangle CMB$  and  $\triangle RNQ$ :  $\frac{MB}{NQ} = \frac{BC}{QR}$  and the included angles  $\angle B = \angle Q$ . **So**  $\triangle CMB \sim \triangle RNQ$  (**SAS**).

9. In  $\triangle ABC$  and  $\triangle DEF$ ,  $\angle B = \angle E$ ,  $\angle C = \angle F$  and  $AB = 3DE$ . Are the two triangles similar, congruent, both or neither? **CBSE 2025, 1 mark**

**SOLUTION**

Two pairs of angles are equal, so the triangles are similar by the AA criterion.

$AB = 3DE$ , so the corresponding sides are not equal. The triangles are therefore not congruent.

**Answer: similar but not congruent**

10.  $\triangle ABC \sim \triangle PQR$  with  $AB = 6$  cm,  $BC = 4$  cm,  $AC = 8$  cm and  $PR = 6$  cm. Find  $PQ + QR$ .

**CBSE 2025, 2 marks**

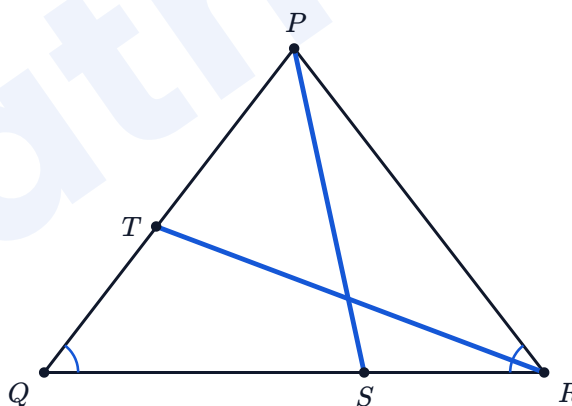
**SOLUTION**

Corresponding sides are in the same ratio:  $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{8}{6} = \frac{4}{3}$

$$PQ = 6 \times \frac{3}{4} = 4.5 \text{ cm and } QR = 4 \times \frac{3}{4} = 3 \text{ cm}$$

**Answer:  $PQ + QR = 7.5$  cm**

11. In  $\triangle PQR$ ,  $S$  is a point on  $QR$  and  $T$  is a point on  $PQ$ . It is given that  $\frac{QR}{QS} = \frac{QT}{PR}$  and  $\angle PQR = \angle PRQ$ . Prove that  $\triangle PQS \sim \triangle TQR$ . **CBSE 2025, 2 marks**



**SOLUTION**

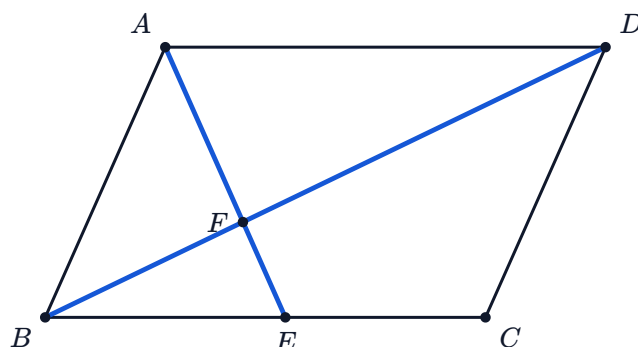
$\angle PQR = \angle PRQ$ , so the sides opposite these angles are equal:  $PR = PQ$ .

Put  $PQ$  in place of  $PR$  in the given relation:  $\frac{QR}{QS} = \frac{QT}{PQ}$ , that is  $\frac{QT}{QR} = \frac{PQ}{QS}$ .

In  $\triangle PQS$  and  $\triangle TQR$ :  $\frac{PQ}{TQ} = \frac{QS}{QR}$  and the included angle  $\angle Q$  is common.

**So  $\triangle PQS \sim \triangle TQR$  by the SAS criterion.**

- 12.**  $ABCD$  is a parallelogram and  $E$  is a point on  $BC$ . The diagonal  $BD$  meets  $AE$  at  $F$ . Prove that  $DF \times EF = FB \times FA$ . **CBSE 2025, 5 marks**



**SOLUTION**

In  $\triangle DFA$  and  $\triangle BFE$ :  $\angle DFA = \angle BFE$  (vertically opposite angles).

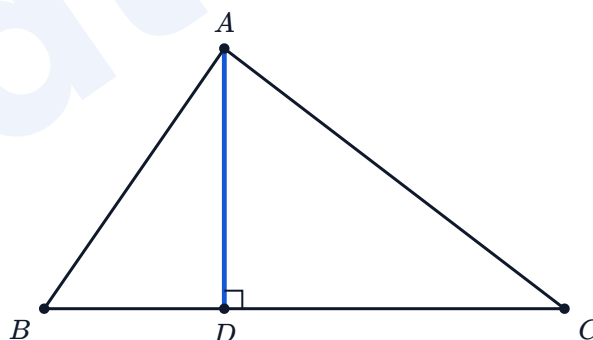
$AD \parallel BC$  and  $BD$  is a transversal, so  $\angle ADF = \angle EBF$  (alternate angles).

So  $\triangle DFA \sim \triangle BFE$  by the AA criterion.

Corresponding sides are proportional:  $\frac{DF}{BF} = \frac{FA}{FE}$ .

Cross-multiplying,  $DF \times EF = FB \times FA$ . **Proved.**

- 13.** In  $\triangle ABC$ ,  $D$  is a point on  $BC$  with  $AD \perp BC$  and  $AD^2 = BD \times DC$ . Prove that  $\angle BAC = 90^\circ$ . **CBSE 2025, 5 marks**



**SOLUTION**

In right triangle  $ADB$ :  $AB^2 = AD^2 + BD^2$  ... (1)

In right triangle  $ADC$ :  $AC^2 = AD^2 + DC^2$  ... (2)

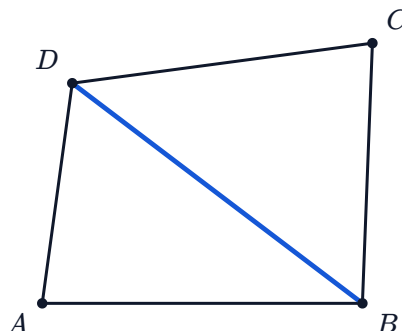
Adding (1) and (2):  $AB^2 + AC^2 = 2AD^2 + BD^2 + DC^2$

Since  $AD^2 = BD \times DC$ :  $AB^2 + AC^2 = BD^2 + 2BD \times DC + DC^2 = (BD + DC)^2 = BC^2$

By the converse of the Pythagoras theorem, the angle opposite  $BC$  is a right angle.

**So  $\angle BAC = 90^\circ$ . Proved.**

- 14.** In quadrilateral  $ABCD$ , the diagonal  $BD$  bisects both  $\angle B$  and  $\angle D$ . Prove that (i)  $\triangle ABD \sim \triangle CBD$  (ii)  $AB = BC$ . **CBSE 2024, 2 marks**



**SOLUTION**

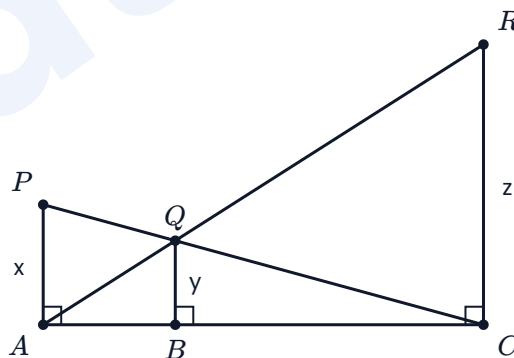
(i) In  $\triangle ABD$  and  $\triangle CBD$ :  $\angle ABD = \angle CBD$  ( $BD$  bisects  $\angle B$ ) and  $\angle ADB = \angle CDB$  ( $BD$  bisects  $\angle D$ ).

So  $\triangle ABD \sim \triangle CBD$  by the AA criterion.

(ii) Corresponding sides are proportional:  $\frac{AB}{CB} = \frac{BD}{BD} = 1$

**Hence**  $AB = BC$ .

- 15.**  $PA$ ,  $QB$  and  $RC$  are all perpendicular to the line  $AC$ , where  $Q$  is the point in which  $PC$  and  $AR$  meet. If  $AP = x$ ,  $BQ = y$  and  $CR = z$ , prove that  $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$ . **CBSE 2024, 5 marks**



**SOLUTION**

In  $\triangle CQB$  and  $\triangle CPA$ :  $\angle C$  is common and  $\angle CBQ = \angle CAP = 90^\circ$ , so  $\triangle CQB \sim \triangle CPA$  (AA).

Hence  $\frac{BQ}{AP} = \frac{BC}{AC}$ , that is  $\frac{y}{x} = \frac{BC}{AC}$ .

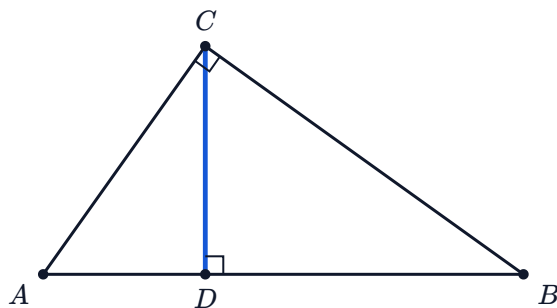
In  $\triangle AQB$  and  $\triangle ARC$ :  $\angle A$  is common and  $\angle ABQ = \angle ACR = 90^\circ$ , so  $\triangle AQB \sim \triangle ARC$  (AA).

Hence  $\frac{BQ}{CR} = \frac{AB}{AC}$ , that is  $\frac{y}{z} = \frac{AB}{AC}$ .

Adding:  $\frac{y}{x} + \frac{y}{z} = \frac{BC + AB}{AC} = \frac{AC}{AC} = 1$

Dividing by  $y$ :  $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$ .

16. In  $\triangle ABC$ ,  $\angle ACB = 90^\circ$  and  $CD \perp AB$ . Prove that  $CD^2 = BD \times AD$ . **CBSE 2019, 3 marks**



**SOLUTION**

In  $\triangle ADC$  and  $\triangle CDB$ :  $\angle ADC = \angle CDB = 90^\circ$ .

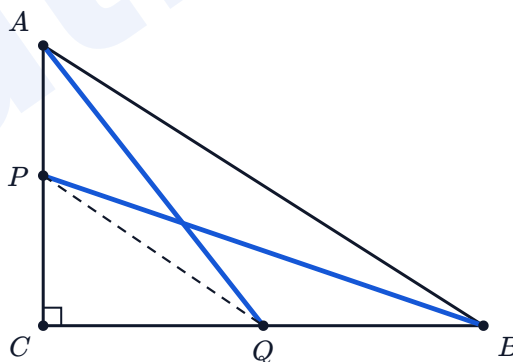
$\angle DAC = 90^\circ - \angle ACD$  (angle sum in  $\triangle ADC$ ) and  $\angle DCB = 90^\circ - \angle ACD$  (because  $\angle ACB = 90^\circ$ ). So  $\angle DAC = \angle DCB$ .

Hence  $\triangle ADC \sim \triangle CDB$  by the AA criterion.

$$\frac{AD}{CD} = \frac{CD}{BD}$$

So  $CD^2 = BD \times AD$ . **Proved.**

17.  $\triangle ABC$  is right-angled at  $C$ .  $P$  is a point on  $CA$  and  $Q$  is a point on  $CB$ . Prove that  $AQ^2 + BP^2 = AB^2 + PQ^2$ . **CBSE 2019, 3 marks**



**SOLUTION**

Each of the triangles  $ACQ$ ,  $PCB$ ,  $ACB$  and  $PCQ$  has a right angle at  $C$ . Apply the Pythagoras theorem to each.

$$AQ^2 = AC^2 + CQ^2 \text{ and } BP^2 = PC^2 + BC^2$$

$$\text{Adding: } AQ^2 + BP^2 = (AC^2 + BC^2) + (PC^2 + CQ^2)$$

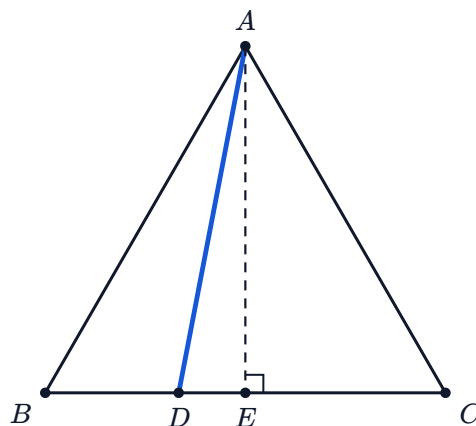
$$AC^2 + BC^2 = AB^2 \text{ and } PC^2 + CQ^2 = PQ^2$$

So  $AQ^2 + BP^2 = AB^2 + PQ^2$ . **Proved.**

### 3 Right-angled triangles

18.  $ABC$  is an equilateral triangle and  $D$  is a point on  $BC$  with  $BD = \frac{1}{3}BC$ . Prove that  $9AD^2 = 7AB^2$ .

CBSE 2018, 4 marks



#### SOLUTION

Let each side be  $a$ . Draw  $AE \perp BC$ . In an equilateral triangle the altitude bisects the base, so  $BE = \frac{a}{2}$ .

$$BD = \frac{a}{3}, \text{ so } DE = BE - BD = \frac{a}{2} - \frac{a}{3} = \frac{a}{6}.$$

$$\text{In right triangle } AEB: AE^2 = a^2 - \frac{a^2}{4} = \frac{3a^2}{4}$$

$$\text{In right triangle } AED: AD^2 = AE^2 + DE^2 = \frac{3a^2}{4} + \frac{a^2}{36} = \frac{28a^2}{36} = \frac{7a^2}{9}$$

**So  $9AD^2 = 7a^2 = 7AB^2$ . Proved.**

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